Model Summary

Data level

$$y_{ijk} \sim \text{Bernoulli}\left(p_{ijk}\right)$$
$$p_{ijk} = \Phi\big(\beta_j\left(s_{ijk} + \theta_{ij}\right)\big)$$

Measurement level

$$\gamma_i = \mu + \Lambda\,\eta_i + \varepsilon_i$$
$$\gamma_i = \left(\theta_{i1}, \theta_{i2}, \theta_{i3}, \theta_{i4}, \theta_{i5}, \theta_{i6}, c_{i1}, c_{i2}, c_{i3}, c_{i4}\right)'$$
$$\Lambda = \begin{pmatrix} \omega_1\lambda_1 & \omega_1\lambda_2 & \frac{\omega_1\lambda_3}{\sqrt{2}} & \frac{\omega_1\lambda_3}{\sqrt{6}} & 0 \\ \omega_2\lambda_1 & \omega_2\lambda_2 & -\frac{\omega_2\lambda_3}{\sqrt{2}} & \frac{\omega_2\lambda_3}{\sqrt{6}} & 0 \\ \omega_3\lambda_1 & \omega_3\lambda_2 & 0 & -\frac{2\omega_3\lambda_3}{\sqrt{6}} & 0 \\ \omega_4\lambda_1 & -\omega_4\lambda_2 & \frac{\omega_4\lambda_3}{\sqrt{2}} & \frac{\omega_4\lambda_3}{\sqrt{6}} & 0 \\ \omega_5\lambda_1 & -\omega_5\lambda_2 & -\frac{\omega_5\lambda_3}{\sqrt{2}} & \frac{\omega_5\lambda_3}{\sqrt{6}} & 0 \\ \omega_6\lambda_1 & -\omega_6\lambda_2 & 0 & -\frac{2\omega_6\lambda_3}{\sqrt{6}} & 0 \\ 0 & 0 & 0 & 0 & \kappa_1 \\ 0 & 0 & 0 & 0 & \kappa_2 \\ 0 & 0 & 0 & 0 & \kappa_3 \\ 0 & 0 & 0 & 0 & \kappa_4 \end{pmatrix}, \quad \eta_i = \begin{pmatrix} \eta_{i1} \\ \eta_{i2} \\ \eta_{i3} \\ \eta_{i4} \\ \eta_{i5} \end{pmatrix}, \quad \varepsilon_i \sim N(\mathbf{0},\, \text{diag}(\delta_1^2, \ldots, \delta_{10}^2))$$
$$\omega_j = \exp(a_j), \quad \sum_{j=1}^{6} a_j = 0, \qquad j = 1, \ldots, 6$$

Structural level

$$\eta_i = \mathbf{B}\,\eta_i + \zeta_i, \qquad \zeta_i \sim N(\mathbf{0},\, \mathbf{I}_5)$$
$$\mathbf{B} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ b_1 & b_2 & b_3 & b_4 & 0 \end{pmatrix}$$
$$\text{Cov}(\eta_i) = \begin{pmatrix} 1 & 0 & 0 & 0 & b_1 \\ 0 & 1 & 0 & 0 & b_2 \\ 0 & 0 & 1 & 0 & b_3 \\ 0 & 0 & 0 & 1 & b_4 \\ b_1 & b_2 & b_3 & b_4 & 1 + b_1^2 + b_2^2 + b_3^2 + b_4^2 \end{pmatrix}$$

Priors

$$
\begin{aligned}
\beta_j &\sim N(0,\, 5^2), && j = 1, \dots, 6 \\
\mu_t &\sim N(0,\, 5^2), && t = 1, \dots, 10 \\
a_j &\sim N(0,\, 1), && j = 1, \dots, 5 \quad (a_6 = -\textstyle\sum_{j=1}^{5} a_j) \\
\lambda_1,\, \lambda_2,\, \lambda_3 &\sim \text{Half-}N(0,\, 1) \\
\kappa_k &\sim \text{Half-}N(0,\, 1), && k = 1, \dots, 4 \\
\delta_j &\sim \text{Half-}N(0,\, 1), && j = 1, \dots, 10 \\
b_l &\sim N(0,\, 1), && l = 1, \dots, 4
\end{aligned}
$$

Variance decomposition of perceptual task j

$$
\mathrm{Var}(\theta_{ij}) = \underbrace{\omega_j^2 \lambda_1^2}_{\text{General}} + \underbrace{\omega_j^2 \lambda_2^2}_{\text{Masking}} + \underbrace{\tfrac{2}{3}\,\omega_j^2 \lambda_3^2}_{\text{Stimulus}} + \underbrace{\delta_j^2}_{\text{Unique}}
$$

$$
\mathrm{General}_j = \frac{\omega_j^2 \lambda_1^2}{\mathrm{tot}_j}, \qquad \mathrm{Masking}_j = \frac{\omega_j^2 \lambda_2^2}{\mathrm{tot}_j}
$$

$$
\mathrm{Stimulus}_j = \frac{\frac{2}{3}\,\omega_j^2 \lambda_3^2}{\mathrm{tot}_j}, \qquad \mathrm{Unique}_j = \frac{\delta_j^2}{\mathrm{tot}_j}
$$

Relation to cognitive performance

$$
v = \mathrm{Var}(\eta_{i5}) = 1 + b_1^2 + b_2^2 + b_3^2 + b_4^2
$$

$$
r_l = \mathrm{Corr}(\eta_{il},\, \eta_{i5}) = \frac{b_l}{\sqrt{v}}, \qquad l = 1, \dots, 4
$$

$$
R_G^2 = r_1^2 = \frac{b_1^2}{v}, \qquad R_M^2 = r_2^2 = \frac{b_2^2}{v}, \qquad R_S^2 = r_3^2 + r_4^2 = \frac{b_3^2 + b_4^2}{v}
$$

$$
r_{\text{fast}} = \mathrm{Corr}(\lambda_1 \eta_{i1} + \lambda_2 \eta_{i2},\, \eta_{i5}) = \frac{\lambda_1 b_1 + \lambda_2 b_2}{\sqrt{(\lambda_1^2 + \lambda_2^2)\, v}}
$$

$$
r_{\text{sharp}} = \mathrm{Corr}(\lambda_1 \eta_{i1} - \lambda_2 \eta_{i2},\, \eta_{i5}) = \frac{\lambda_1 b_1 - \lambda_2 b_2}{\sqrt{(\lambda_1^2 + \lambda_2^2)\, v}}
$$

$$
r_a = \mathrm{Corr}\left(\lambda_1 \eta_{i1} + \lambda_3 (h_{a1} \eta_{i3} + h_{a2} \eta_{i4}),\, \eta_{i5}\right) = \frac{\lambda_1 b_1 + \lambda_3 \left(h_{a1} b_3 + h_{a2} b_4\right)}{\sqrt{\left(\lambda_1^2 + \frac{2}{3}\lambda_3^2\right) v}}, \qquad a \in \{\text{Pi, Letter, Gabor}\}
$$

Result1: Covaraince matrix

Result2: Variance decomposition

Task	ω_j	General	Masking	Stimulus	Unique
TPi	1.64 [1.22, 2.10]	0.47 [0.28, 0.66]	0.04 [0.00, 0.10]	0.20 [0.11, 0.33]	0.28 [0.01, 0.56]
TLetter	0.74 [0.51, 0.99]	0.33 [0.15, 0.53]	0.03 [0.00, 0.08]	0.14 [0.06, 0.26]	0.50 [0.20, 0.77]
TGabor	0.90 [0.59, 1.23]	0.34 [0.14, 0.56]	0.03 [0.00, 0.08]	0.15 [0.06, 0.28]	0.49 [0.13, 0.79]
SPi	1.97 [1.52, 2.43]	0.54 [0.34, 0.71]	0.05 [0.00, 0.11]	0.23 [0.13, 0.35]	0.18 [0.00, 0.46]
SLetter	0.92 [0.69, 1.14]	0.56 [0.33, 0.73]	0.05 [0.00, 0.11]	0.24 [0.13, 0.36]	0.14 [0.00, 0.48]
SGabor	0.54 [0.38, 0.71]	0.49 [0.25, 0.70]	0.04 [0.00, 0.10]	0.21 [0.10, 0.34]	0.25 [0.00, 0.59]
Average		**0.46 [0.36, 0.55]**	**0.04 [0.00, 0.09]**	**0.20 [0.12, 0.28]**	**0.31 [0.22, 0.42]**

Result3: Contribution to cognitive performance

Factor	b	BF_{10}	r
G (average level)	1.01 [0.50, 1.72]	9,731.86	0.59 [0.37, 0.77]
M (Temporal − Spatial)	−0.34 [−1.16, 0.40]	0.55	−0.19 [−0.55, 0.24]
S1 (Pi − Letter)	0.39 [−0.05, 0.97]	0.957	0.23 [−0.03, 0.48]
S2 (Pi and Letter vs Gabor)	−0.47 [−1.19, 0.07]	1.01	−0.27 [−0.56, 0.05]

Composite	r with cognition
Temporal tasks	0.51 [0.24, 0.74]
Spatial tasks	0.62 [0.40, 0.79]
Pi tasks	0.53 [0.29, 0.73]
Letter tasks	0.31 [0.02, 0.57]
Gabor tasks	0.64 [0.41, 0.83]